FSAs - Solution

1. **2002-03**

Solution: b, d and e

a. 101101110 – is missing a final 1
b. 1*0(10*1U01) 01*(110*1U01*00) – is the regular expression of the FSA
c. 101010101010 – is only correct for the first 10 digits
d. 00100100 – is correct along the bottom path
e 1*010*101*110*1 – is correct along the top path

2. **2003-04**

Solution: 2

The shortest string with 6 1's is : 1110111 (lower-top path)
The longest string with 6 1's is: 110101011 (lower-bottom)
9 – 7 = 2 is the difference between the longest and the shortest length.

3. **2004-05**

Solution: As shown

The regular expression is: a[(ba*a (ba ∪ ab) a) ∪ (aba*ab)]**.** Spacing doesn't matter.

4. **2005-06**

Solution: D

Path D is the only one that is accepted.
It travels the upper branch of the lower path of the FSA.

5. **2006-07**

Solution: III

III is the only regular expression accepted using the lower branch.

6. **2008-09**

Solution: a, b, e

c and f are not accepted because there cannot be a decimal and an E in the same expression.
d is incorrect there must be a digit before the decimal point.

7. **2009-10**

Solution: a, c, e, f

The regular expression for this FSA is: (00*1 ∪ 10)1*((010* ∪ 1(01*1 ∪ 11)0*)
a, c, e and f are accepted by the FSA. b fails at the final 1. d never gets out of the first OR

8. 2012-13

Solution: b, c, e, f

The FSA translates to the following regular expression:

101*0 ((11*0 (001 U 11*00) 0*0) U (00*11* (011*0 U 110*1) 1*1)

a fails on the very top branch. d fails on the top branch of the bottom branch.

9. **2013-14**

Solution: a, c, d, f

The FSA translates to the following regular expression:

11*0 ((10*(010*1 U 110*1)01*) U (11*0(10*11 U 10*0)01*)) 01* (111*1 U 10*00)

b fails at the 4th 0. e fails at the 2nd 0.

10. 2014-15

Solution:

The FSA translates to the following regular expression:

10*00* (101*1 U 11*01*1) 01*01